Winston Wang

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EXPERIENCE

Monclus Vending Services LLC **Mar 2026 – Present**
Contract Engineer
- Design and manufacturing of custom components for mechanical assemblies, including discontinued parts
- Application of GD&T and Tolerance Stack Up Analysis to ensure proper fitting of parts with multiple dependencies

AEM **Dec 2024 – Jul 2025**
Prototype Engineer
- Co-designed a hybrid-electric VTOL aircraft for fuel-efficient operations within a space-constrained environment
- Produced a functional small-scale demonstrator for technology commercialization

Entrepreneurship and Technology Innovation Center (ETIC) **May 2023 – May 2025**
Lead Engineer & Project Manager
- Led the development of aviation-focused prototypes, including VTOL drones, 3D-motion pedal control systems, and low visibility flight training goggles, enabling the lab's entry into the aerospace industry
- Lead engineer overseeing mechanical part design, utilizing CAD and manufacturing for rapid prototyping
- Developed prototype technologies from NASA patents, integrating mechanical, electrical, and software components across multi-disciplinary teams

SKILLS

Technical: Sensor & Actuator Integration, GD&T, Additive & Subtractive Manufacturing, Engineering Analysis
Analytical/Communication: Root Cause Analysis, Technology Commercialization, PDR-CDR, Technical Documentation
Software/Languages: Onshape, Solidworks, MATLAB/Simulink, ANSYS, Python, LabVIEW, KiCad

PROJECTS

Counter Rotating Coaxial IBC System **In Progress**
- Physical demonstrator featuring twin counter-rotating coaxial helicopter blades with Individual Blade Control system.
- Designed electro-mechanical model via FEA and CFD, along with vibration sensors and wind tunnel testing.

System, Apparatus and Method for Pedal Control **May 2025**
- US Patent #10,180,699 by NASA's Johnson Space Center for controlling movement of an object in three-dimensional space
- Built a 6-DOF free-movement pedal mechanism with sensor-integrated motion tracking, using Raspberry Pi and Python for real-time control input translation to a simulation

Variable Visibility Glasses for Instrument Flight Training **Sep 2024**
- US Patent #8,411,214 by NASA's Langley Research Center for a flight training system to selectively restrict vision
- Developed head-tracking IR sensor system stabilized within a cockpit, and engineered a lightweight head-mounted device featuring PDLC glass, sensors, and embedded computing unit

VTOL Dragonfly **May 2024**
- Designed a VTOL UAV for autonomous delivery and search-and-rescue, featuring a tiltrotor system, retractable wings, modular cargo pod, and onboard machine learning camera for target identification
- Optimized aerodynamic profile, delivery module, and propulsion configuration through iterative testing

Gamified Silk Screen Cleaner (NY State 1st Place) **Apr 2024**
- Partnered with NYSID to develop a safe and efficient plastisol ink removal system, supporting workers with disabilities
- Designed a high-torque cleaning mechanism with aluminum-acrylic enclosure, reducing chemical exposure

EDUCATION

New York Institute of Technology **May 2026**
Bachelor of Science in Mechanical Engineering, Aerospace Concentration
Relevant Coursework: Advanced Aerodynamics, Advanced Propulsions, Flight Vehicle Design, Modern Manufacturing

AWARDS & CERTIFICATIONS

NASA T2X Certification of Excellence for Prototype Development: Recognition of outstanding work in prototype development using NASA-patented technologies
1st place NYSID CREATE: Placed first amongst NY state colleges for assistive technology innovations
CoECS Innovator of the Year: Recognition of innovation and entrepreneurship for development of a new technology
SOLIDWORKS CSWA Certification: Proficiency in 3D modeling, part design, and assemblies using SolidWorks